COMSATS University Islamabad

Attock Campus

Department Of Computer Science

Course	*Info. Security (LAB)*
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Diffie-Hellman protocol:

Code:

Welcome | Diffe Helman Lab Task 20 April.py

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```python
#Diffie-Hellman Code
#Power function to return value of (a^b) mod p
def power(a, b, p):
    return pow(a, b, p)

def main():

    P = 23
    print("The value of P:", P)

    G = 9
    print("The value of G:", G)

    a = 4
    print("The private key a for Alice:", a)

    x = power(G, a, P)

    b = 3
    print("The private key b for Bob:", b)

    y = power(G, b, P)

    ka = power(y, a, P)  # Alice computes  secret key
    kb = power(x, b, P)  # Bob computes  secret key

    print("Secret key for Alice is:", ka)
    print("Secret key for Bob is:", kb)

if __name__ == "__main__":
    main()
```

Output:

```
PS E:\Semester 4\Info Sec Lab>  & 'c:\Users\LENOVO\AppData\Local
8.0-win32-x64\bundled\libs\debugpy\launcher' '50548' '--' 'e:\Se
The value of P: 23
The value of G: 9
The private key a for Alice: 4
The private key b for Bob: 3
Secret key for Alice is: 9
Secret key for Bob is: 9
PS E:\Semester 4\Info Sec Lab>
```

Graded Home Task

Search about latest attack on key exchange algorithms and provide its implementation in python.

Code:

Welcome | 1.py | Diffe Helman Lab Task 20 April Task 2.py

Semester 4 > Info Sec Lab > Diffe Helman Lab Task 20 April Task 2.py > ...

```python
import time
def power(a, b, p):
    return pow(a, b, p)

P = 23
G = 9
private_key = 6

def server_compute(client_public):
    return power(client_public, private_key, P)

print("Server Ready...\n")

for i in range(1, 10):
    fake_key = i * 99999999   # attacker sends large/fake values
    start = time.time()

    shared = server_compute(fake_key)

    end = time.time()
    print(f"Request {i}: Computation Time = {end - start:.6f} sec")
```

Output:

```
PS E:\Semester 4\Info Sec Lab>  & 'c:\Users\LENOVO\AppData\
8.0-win32-x64\bundled\libs\debugpy\launcher' '50868' '--' '
Server Ready...

Request 1: Computation Time = 0.000009 sec
Request 2: Computation Time = 0.000009 sec
Request 3: Computation Time = 0.000004 sec
Request 4: Computation Time = 0.000003 sec
Request 5: Computation Time = 0.000005 sec
Request 6: Computation Time = 0.000007 sec
Request 7: Computation Time = 0.000004 sec
Request 8: Computation Time = 0.000004 sec
Request 9: Computation Time = 0.000005 sec
PS E:\Semester 4\Info Sec Lab>
```

0 0 Update is ready, click to restart.